

E067

Calculus I

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A sensor model logged every 0.5 seconds is $f(x) = (x^2 - 9)/(x - 3)$. Plugging in $x = 3$ gives $0/0$. The sensor was calibrated last month. What is the limit of $f(x)$ as x approaches 3?

- A. 3
- B. 6
- C. The limit does not exist because $f(3)$ is undefined.
- D. 0

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

An analyst who joined in March models cost as $C(x) = 4x^3 - 5x + 7$ dollars, where x is thousands of units. How fast is cost changing when $x = 2$?

- A. 43 dollars per thousand units
- B. 29
- C. 20
- D. 48

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A bacterial culture in a lab that runs 24 hours a day grows as $y = e^{(3x)}$. What is dy/dx at $x = 0$?

- A. 0
- B. 1
- C. 3
- D. $1/3$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

What is the derivative of $\sin(x^2)$?

- A. $2x \cdot \cos(x^2)$
- B. $\cos(2x)$
- C. $-2x \cdot \cos(x^2)$
- D. $\cos(x^2)$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

On a test rig that cost \$80,000, an engineer models power as $P(t) = (t^2 + 1) \cdot t^3$. What is $P'(1)$?

- A. -4
- B. 8
- C. 2
- D. 6

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A company with 14 employees has profit $P(x) = -2x^2 + 80x - 300$ dollars for x units. How many units maximize profit?

- A. 40 units
- B. 500
- C. 300 units
- D. 20 units

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A field technician without a calculator estimates $\sqrt{26}$ using $f(x) = \sqrt{x}$ near $x = 25$. The cable reel beside her weighs 40 kg. What is the linear estimate?

- A. 5.2
- B. 4.9
- C. 5
- D. 5.1

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

The linearization of f at $a = 1$ is $L(x) = 3 + 2(x - 1)$. What are $f(1)$ and $f'(1)$?

- A. $f(1) = 5$ and $f'(1) = 2$
- B. $f(1) = 1$ and $f'(1) = 3$
- C. $f(1) = 3$ and $f'(1) = 2$
- D. $f(1) = 2$ and $f'(1) = 3$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Water flows into a tank at $r(t) = 2t$ liters per minute. The tank is painted blue and holds up to 500 liters. How much water enters between $t = 1$ and $t = 3$ minutes?

- A. -8 liters
- B. 9 liters
- C. 8 liters
- D. 4 liters

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Which is the general antiderivative of $6x^2$?

- A. $12x$
- B. $2x^3 + C$
- C. $6x^3 + C$
- D. $2x^3$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Requests to a startup's API grow at a rate proportional to their number: $dy/dt = 0.05y$, with $y = 1,000$ requests per hour at $t = 0$ days. The startup has a single data center. About how many requests per hour will there be at $t = 20$ days?

- A. About 2,653
- B. 2,000
- C. About 1,051
- D. About 2,718

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A quantity follows $y = 500e^{(kt)}$ and doubles every 10 hours. What is k ?

- A. About 0.0693 per hour
- B. About 6.93 per hour
- C. 0.1 per hour
- D. 0.2 per hour

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A sensor model ~~logged every 0.5 seconds~~ is $f(x) = (x^2 - 9)/(x - 3)$. Plugging in $x = 3$ gives $0/0$. ~~The sensor was calibrated last month.~~ What is the limit of $f(x)$ as x approaches 3?

Target word: limit at a hole **Decides:** $(x^2 - 9)/(x - 3) = x + 3$ for $x \neq 3$.

A	Wrong stem number	The x-value, not the limit.
B	Correct	Cancel $(x - 3)$; $x + 3 \rightarrow 6$.
C	Confusion set	Mixes up the function value with the limit.
D	Dropped element	Reads $0/0$ as 0 and skips simplifying.

2. Answer A Concept 2

~~An analyst who joined in March~~ models cost as $C(x) = 4x^3 - 5x + 7$ dollars, where x is thousands of units. How fast is cost changing when $x = 2$?

Target word: instantaneous rate at $x = 2$ **Decides:** $C'(x) = 12x^2 - 5$.

A	Correct	$12(4) - 5 = 43$.
B	Adjacent category	That is $C(2)$, the cost, not its rate.
C	Inverted operation	Evaluates an antiderivative instead of the derivative.
D	Dropped element	Drops the -5 term's derivative.

3. Answer C Concept 3

A bacterial culture ~~in a lab that runs 24 hours a day~~ grows as $y = e^{(3x)}$. What is dy/dx at $x = 0$?

Target word: derivative using the chain rule **Decides:** $dy/dx = 3e^{(3x)}$.

A	Confusion set	Applies the power rule $(3x \cdot e^{(3x-1)})$ to an exponential.
B	Dropped element	Forgets the inner derivative 3.
C	Correct	$3e^0 = 3$.
D	Inverted operation	Divides by 3 as if integrating.

4. Answer A Concept 3 · transfer

What is the derivative of $\sin(x^2)$?

Target word: chain rule with a trig outer function **Decides:** Outer derivative $\cos(x^2)$ times inner derivative $2x$.

A	Correct	cos of the inside times $2x$.
B	Confusion set	Differentiates the inside and puts it inside the cosine.
C	Inverted operation	The derivative of sine has no minus sign; that is cosine's derivative.
D	Dropped element	Leaves out the inner derivative.

5. Answer B Concept 4

On a test rig that cost \$80,000, an engineer models power as $P(t) = (t^2 + 1) \cdot t^3$. What is $P'(1)$?

Target word: derivative of a product **Decides:** $u = t^2 + 1$, $v = t^3$: $u'v + uv'$.

A	Inverted operation	Subtracts the terms, as in a quotient rule.
B	Correct	$2t \cdot t^3 + (t^2 + 1) \cdot 3t^2 = 2 + 6 = 8$.
C	Dropped element	Keeps only the $u'v$ term.
D	Confusion set	$u' \cdot v' = 2t \cdot 3t^2$ multiplies the derivatives.

6. Answer D Concept 5

A company with 14 employees has profit $P(x) = -2x^2 + 80x - 300$ dollars for x units. How many units maximize profit?

Target word: x at the maximum **Decides:** $P'(x) = -4x + 80 = 0$.

A	Dropped element	Uses $-2x + 80 = 0$, dropping the power rule's factor of 2.
B	Adjacent category	That is the maximum profit $P(20)$, not the number of units.
C	Wrong stem number	The fixed-cost figure in the stem.
D	Correct	$x = 80/4 = 20$; $P'' = -4 < 0$, so it is a maximum.

7. Answer D Concept 6

A field technician without a calculator estimates $\sqrt{26}$ using $f(x) = \sqrt{x}$ near $x = 25$. The cable reel beside her weighs 40 kg. What is the linear estimate?

Target word: linear approximation value **Decides:** $f(25) = 5$, $f'(25) = 1/(2 \cdot 5) = 0.1$.

A	Confusion set	Uses $1/\sqrt{x}$ instead of $1/(2\sqrt{x})$ for the derivative.
B	Inverted operation	Subtracts the correction instead of adding it.
C	Dropped element	Uses $f(a)$ only, no slope term.
D	Correct	$5 + 0.1(26 - 25) = 5.1$.

8. Answer C Concept 6 • transfer

The linearization of f at $a = 1$ is $L(x) = 3 + 2(x - 1)$. What are $f(1)$ and $f'(1)$?

Target word: read $f(a)$ and $f'(a)$ back from L **Decides:** $L(x) = f(a) + f'(a)(x - a)$.

A	Adjacent category	5 is $L(2)$, the estimate at $x = 2$.
B	Confusion set	Treats $a = 1$ as the function value.
C	Correct	Constant term is $f(a)$; coefficient of $(x - a)$ is $f'(a)$.
D	Inverted operation	Swaps the two parts.

9. Answer C Concept 7

Water flows into a tank at $r(t) = 2t$ liters per minute. ~~The tank is painted blue and holds up to 500 liters.~~ How much water enters between $t = 1$ and $t = 3$ minutes?

Target word: accumulated amount **Decides:** Antiderivative t^2 ; evaluate $3^2 - 1^2$.

A	Inverted operation	Computes $F(1) - F(3)$.
B	Dropped element	Uses $F(3)$ and drops $F(1)$.
C	Correct	$F(3) - F(1) = 9 - 1 = 8$.
D	Confusion set	$r(3) - r(1)$ subtracts rates, not accumulated amounts.

10. Answer B Concept 7 • transfer

Which is the general antiderivative of $6x^2$?

Target word: indefinite antiderivative **Decides:** Raise the power, divide by the new power, add C.

A	Inverted operation	That is the derivative.
B	Correct	$6x^3/3 + C$.
C	Confusion set	Raises the power but forgets to divide by 3.
D	Dropped element	Missing $+ C$.

11. Answer D Concept 8

Requests to a startup's API grow at a rate proportional to their number: $dy/dt = 0.05y$, with $y = 1,000$ requests per hour at $t = 0$ days. ~~The startup has a single data center.~~ About how many requests per hour will there be at $t = 20$ days?

Target word: solve $dy/dt = ky$ **Decides:** $y = 1,000 \cdot e^{(0.05 \cdot 20)} = 1,000e$.

A	Adjacent category	$1,000(1.05)^{20}$ is discrete daily compounding, not continuous growth.
B	Confusion set	Linear growth: $1,000 + 0.05 \times 1,000 \times 20$.
C	Dropped element	$1,000e^{0.05}$ leaves out t .
D	Correct	$1,000 \times e^1 \approx 2,718$.

12. Answer A Concept 8 • transfer

A quantity follows $y = 500e^{(kt)}$ and doubles every 10 hours. What is k ?

Target word: solve for the rate constant **Decides:** $e^{(10k)} = 2$, so $k = \ln 2 / 10$.

A	Correct	$0.6931 / 10$.
B	Inverted operation	Multiplies $\ln 2$ by 10 instead of dividing.
C	Confusion set	$1 / \text{doubling time}$ is not the continuous rate.
D	Dropped element	$2 / 10$ leaves out the logarithm.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	11	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	0	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Limit vs. function value	A limit describes what $f(x)$ approaches near a point; it can exist even where f is undefined (for example, after canceling a common factor).	Declaring the limit undefined because plugging in gives $0/0$.	A fraction that gives $0/0$ at the point.
2	Power rule and the derivative as a rate	$d/dx x^n = n \cdot x^{n-1}$; $f'(a)$ is the instantaneous rate of change at a , not the value $f(a)$.	Reporting $f(a)$ when the question asks for the rate.	"How fast... at the moment..."
3	Chain rule	$d/dx f(g(x)) = f'(g(x)) \cdot g'(x)$: differentiate the outside, keep the inside, multiply by the inside's derivative.	Forgetting to multiply by the inner derivative.	A function inside another: $e^{(kx)}$, $\sin(x^2)$, $(\dots)^n$.
4	Product rule	$(uv)' = u'v + uv'$.	Multiplying the two derivatives together.	Two x -expressions multiplied together.
5	Optimization	Set $f'(x) = 0$ to find critical points; a negative second derivative there marks a maximum. Answer the quantity asked (x or $f(x)$).	Reporting the maximum value when the question asks where it occurs, or vice versa.	"Maximize," "minimize," "how many units."
6	Linear approximation	Near $x = a$, $f(x) \approx f(a) + f'(a)(x - a)$.	Using $f(a)$ alone, or the wrong derivative.	Estimate a value close to an easy one without a calculator.
7	Antiderivatives and the Fundamental Theorem	$\int_a^b f(x) dx = F(b) - F(a)$ where $F' = f$; an indefinite antiderivative needs $+ C$.	Evaluating f instead of F , or using only $F(b)$.	Accumulated amount from a rate over an interval.
8	Exponential growth differential equation	$dy/dt = ky$ has solution $y = y_0 e^{(kt)}$; doubling time is $\ln 2 / k$.	Treating proportional growth as linear, or dropping the logarithm.	"Grows at a rate proportional to its size."

Course scope (public description): Calculus I introduces the fundamental ideas of limits, derivatives, anti-derivatives, and basic differential equations as tools for modeling and solving real-world problems. Students explore how functions behave, how rates of change are measured and optimized, and how derivatives lead to applications in linear approximations and optimization. The basics of how accumulated quantities (antiderivatives) are computed using the Fundamental Theorem of Calculus are also introduced. Emphasis is placed on polynomial, exponential, logarithmic, and trigonometric functions, with applications drawn from science, engineering, economics, and technology. Students will focus on conceptual understanding, graphical interpretation, and applied problem-solving rather than formal proofs. By the end of the course, students will be able to analyze change, model systems, and apply calculus techniques to practical scenarios.